Stochastic Traffic Signal Timing Optimization

A queuing and Monte Carlo study

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A precise, limited question

- **Decision:** allocate 80 seconds of green time across the North, East, South, and West approaches.
- **Signal operation:** four protected phases in a 96-second cycle, including 16 seconds of clearance.
- **Scenario:** 12 movement queues, initially empty, with assumed Poisson arrivals over a 600-second period.

Objective

Minimize expected cumulative waiting, measured in vehicle-seconds. No field measurements are used in this primary experiment.

From arrivals to waiting cost

Poisson arrivals

Uniform pseudorandom draws are transformed into exponential interarrival gaps:

$$\Delta t = -\frac{\ln(1-U)}{\lambda}.$$

A turning draw assigns each vehicle to a dedicated left, through, or right queue.

Queue-area objective

$$D_T(\mathbf{g}) = \sum_{m=1}^{12} \int_0^T Q_m(t;\mathbf{g})\, dt.$$

This is cumulative waiting in vehicle-seconds, including partial waiting still in progress at the horizon.

Matched comparison

Every candidate timing is scored against the same training arrivals to reduce noise in the search.

Search, then evaluate on fresh traffic

1. Enumerate feasible green allocations on the selected timing grid.
2. Score every candidate on shared training arrivals.
3. Select the candidate with the lowest estimated mean waiting.
4. Evaluate the selected plan on new, matched arrivals.

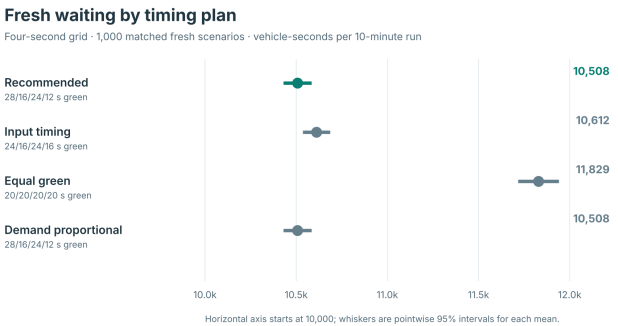
Why simulation?

Service stops at red and uses fixed headways. A continuously serving M/M/1 formula does not represent this signal operation exactly.

Reported measures

Mean waiting, paired differences, queue profiles, residual backlog, and uncertainty intervals.

The ten-minute reference result

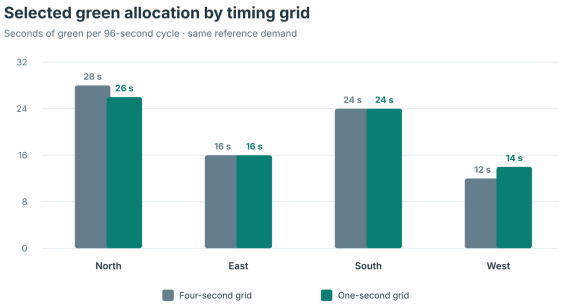


Four-second-grid selection

- Green (N/E/S/W): **28 / 16 / 24 / 12 s**
- Fresh mean: **10,507.8 vehicle-seconds/run**
- Saving vs entered timing: **104.2 vehicle-seconds/run**
- 1,000 fresh runs; paired 95% interval: **69.6–138.8**

The selected plan equals the demand-proportional baseline; the mean reduction is modest.

Fine-grid timing result

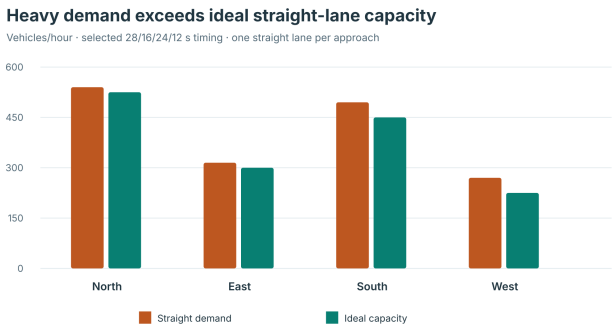


Selected green: 26 / 16 / 24 / 14 s

- 19,369 feasible allocations; 128 training runs per candidate.
- Search runtime: **6.86 seconds**.
- On 5,000 matched fresh runs: **44.8 vehicle-seconds/run** saved (0.42%).
- Paired 95% interval: **34.8–54.8**.

This supports a small difference between the selected plans for this model; it does not establish a global optimum.

Heavy demand can exceed modeled capacity

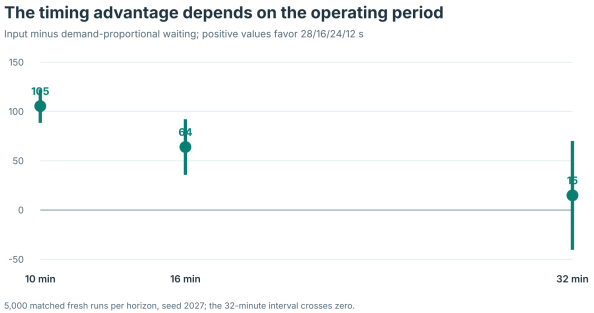


- Heavy-scenario arrival rate: **2,700 vehicles/hour**.
- Straight-stream demand requires **86.4 seconds** of green.
- Only **80 seconds** of green are available per cycle.

Residual queue

Mean backlog is 54.4 vehicles at 10 minutes and 110.1 vehicles at 32 minutes.

Interpret the recommendation within its scope



10-minute horizon

Saving: **105.5 vehicle-seconds/run**
Paired 95% interval: 88.5–122.5.

32-minute horizon

Saving: **15.0 vehicle-seconds/run**
Paired 95% interval: -40.3–70.2; no resolved difference.

Field counts and calibration are needed before applying the timing outside this model.